AP PRECALCULUS

Unit 1 • Lesson 1.11 • Notes (filled in)

Topic 1.11 • Rational Functions and Concept of Limit

Notes (filled in)

Learning goals

- Use limit notation to describe the behavior of rational functions near holes, near vertical asymptotes, and at the ends.
- Rewrite rational expressions (factoring, polynomial division) to evaluate limits and find asymptotes.
- Use the remainder theorem and the binomial theorem to work with equivalent forms of polynomials.

Vocabulary. limit, one-sided limit, equivalent expression, polynomial division, remainder theorem, binomial theorem, Pascal's triangle.

1. Three kinds of limit statements

Key idea

Hole at $x = a$: $\lim_{x \to a} r(x) =$ a real number L ($r(a)$ undefined)

Vertical asymptote $x = a$: $\lim_{x \to a^{\pm}} r(x) = \pm\infty$

Horizontal asymptote $y = b$: $\lim_{x \to \pm\infty} r(x) = b$

Example 1 • Describe everything with limits

$r(x) = \dfrac{2x^2 + 2x - 4}{x^2 - 1}$.

$r(x) = \dfrac{2(x+2)(x-1)}{(x-1)(x+1)}$. Hole: $\lim_{x \to 1} r(x) = \frac{2(3)}{2} = 3$. Asymptote: $\lim_{x \to -1^-} r(x) = -\infty$, $\lim_{x \to -1^+} r(x) = +\infty$ (numerator of $\frac{2(x+2)}{x+1}$ near 2). Ends: $\lim_{x \to \pm\infty} r(x) = 2$.

2. Equivalent forms reveal limits

Key idea

Dividing $p(x)$ by $(x - a)$ gives $p(x) = (x - a)q(x) + R$, and the **remainder theorem** says $R =$ $p(a)$. So $(x - a)$ is a factor exactly when $p(a) = 0$.

Writing $\dfrac{p(x)}{d(x)} = q(x) + \dfrac{R(x)}{d(x)}$ shows that the graph approaches $y = q(x)$ as $x \to \pm\infty$ when $\deg R < \deg d$.

Example 2 • Synthetic division

Divide $2x^3 - 3x^2 - 11x + 6$ by $x - 3$ and factor completely.

3	2	-3	-11	6
		6	9	-6
	2	3	-2	0

Quotient $2x^2 + 3x - 2 = (2x - 1)(x + 2)$, remainder 0.

$2x^3 - 3x^2 - 11x + 6 = (x - 3)(2x - 1)(x + 2)$; zeros 3, $\frac{1}{2}$, -2.

Example 3 • A limit and an asymptote from division

Let $r(x) = \dfrac{x^3 + 2x^2 - 5x + 1}{x - 1}$. Find $\lim_{x \to 1^+} r(x)$ and describe the end behavior.

Synthetic division by 1: $1, 3, -2$, remainder -1: $r(x) = x^2 + 3x - 2 - \dfrac{1}{x - 1}$.
As $x \to 1^+$, $-\frac{1}{x-1} \to -\infty$: $\lim_{x \to 1^+} r(x) = -\infty$. As $x \to \pm\infty$, r behaves like $x^2 + 3x - 2$: $\lim_{x \to \pm\infty} r(x) = \infty$.

3. The binomial theorem

The coefficients of $(a + b)^n$ are row n of Pascal's triangle : row 4 is 1, 4, 6, 4, 1 .

Example 4

Expand $(x - 2)^4$ and $(2x + 1)^3$.

$(x - 2)^4 = x^4 - 8x^3 + 24x^2 - 32x + 16$. $(2x + 1)^3 = 8x^3 + 12x^2 + 6x + 1$.

Common mistakes

- Writing $\lim_{x \to a} r(x) = \infty$ when the one-sided limits have opposite signs.
- Leaving out zero coefficients in synthetic division ($x^3 + 5$ is $1, 0, 0, 5$).

Check your understanding

1. The remainder when $x^4 - 3x + 1$ is divided by $x + 1$ is 5 .
2. $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2} =$ 4 . 3. The coefficient of x^2 in $(x + 3)^3$ is 9 .